Exercițiu 1.

Cerința:

- Să se afle expresia polinomului Bernstein $B_m(f; a, b)$ corespunzător unui interval compact $[a, b]$ și unei funcții f definite pe acest interval.

Rezolvare:

$$B_m(f)(x) = \sum_{k=0}^{m} \binom{m}{k} \cdot x^k (1-x)^{m-k} f\left(\frac{k}{m}\right)$$

Facem schimbarea de variabilă $x = \frac{y-a}{b-a}$ $(x \in [0,1], y \in [a,b])$

$$\Rightarrow B_m(f)(x) = \sum_{k=0}^{m} \binom{m}{k} x^k (1-x)^{m-k} \cdot$$

$$= \sum_{k=0}^{m} \binom{m}{k} \left(\frac{y-a}{b-a}\right)^k \left(1 - \frac{y-a}{b-a}\right)^{m-k} f\left(a + (b-a)\frac{k}{m}\right)$$

$$= \sum_{k=0}^{m} \binom{m}{k} \frac{(y-a)^k}{(b-a)^k} \cdot \frac{(b-y)^{m-k}}{(b-a)^{m-k}} \cdot f\left(a + (b-a)\frac{k}{m}\right)$$

$$= \frac{1}{(b-a)^m} \sum_{k=0}^{m} \binom{m}{k} (y-a)^k (b-y)^{m-k} \cdot f\left(a + \frac{(b-a)k}{m}\right)$$

$$B_m(f)(y; a, b) = \sum_{k=0}^{m} \binom{m}{k} \frac{(y-a)^k \cdot (b-y)^{m-k}}{(b-a)^m} \cdot f\left(a + \frac{(b-a)k}{m}\right)$$

Exercițiu 2:

Cerință:

• Determinați $B_m(f)(x)$ în cazul când $f(x)=e^{\alpha x}$, $\alpha\in\mathbb{R}$

Rezolvare:

$$B_m(f)(x)=\sum_{k=0}^{m}\binom{m}{k}x^k(1-x)^{m-k}f\left(\frac{k}{m}\right)=$$

$$=\sum_{k=0}^{m}\binom{m}{k}x^k(1-x)^{m-k}e^{\alpha\frac{k}{m}}$$

$$=\sum_{k=0}^{m}\binom{m}{k}\left(x e^{\frac{\alpha}{m}}\right)^k(1-x)^{m-k}$$

$$=\left(x e^{\frac{\alpha}{m}}+1-x\right)^m$$

$$\Rightarrow B_m(f)(x)=\left(x e^{\frac{\alpha}{m}}+1-x\right)^m$$

Exercițiul 3

Cerință:

- Să se arate că pentru orice funcție $f \in C^{+}[0,1]$, f convexă avem $f(x) \le B_m(f)(x)$, $(\forall)\ x \in [0,1]$

Rezolvare:

$$f \text{ convexă} \Rightarrow f\left(\sum_{k=0}^{m} \alpha_k x_k\right) \le \sum_{k=0}^{m} \alpha_k f(x_k)$$

$$\alpha_k \in [0,1],\ \sum_{k=0}^{m} \alpha_k = 1$$

$$\Rightarrow f\left(\sum_{k=0}^{m} b_{m,k}(x)\frac{k}{m}\right) \le \sum_{k=0}^{m} b_{m,k}(x) f\left(\frac{k}{m}\right)$$

$$\sum_{k=0}^{m} b_{m,k}(x)\frac{k}{m} = \sum_{k=0}^{m} \frac{m!}{(m-k)!\cdot k!}\cdot\frac{k}{m}\cdot x^k(1-x)^{m-k} =$$

$$= \sum_{k=1}^{m} \frac{m!}{(m-k)!\cdot k!}\,\frac{k}{m}\,x^k(1-x)^{m-k} =$$

$$= \sum_{k=1}^{m} \frac{(m-1)!}{(k-1)!\cdot(m-k)!}\,x^k(1-x)^{m-k} =$$

$$= x\sum_{k=1}^{m} \binom{m-1}{k-1} x^{k-1}(1-x)^{m-k} =$$

$$\text{not: } k-1=j \quad = x\sum_{j=0}^{m-1} \binom{m-1}{j} x^j (1-x)^{m-j-1} =$$

$$= x\,(x+(1-x))^{m-1} = x$$

$$\Rightarrow f(x) \le \sum_{k=0}^{m} b_{m,k}(x) f\left(\frac{k}{m}\right) = B_m(f)(x)$$

$$\Rightarrow f(x) \le B_m(f)(x)$$